

FPGA 6502 Address Decoder Project Service Manual

Revision 1 - Known Good Baseline

This manual documents the current project before any BR2 work. It is intended to be committed to Git and expanded as the project grows.

NASA Design Rules

1. Drive every output.
2. Connect every input.
3. Start with safe defaults.
4. Active-low signals end with _n.
5. Change one thing at a time.
6. Never gate the FPGA clock.
7. Use clear names.
8. Write code that is easy to read.
9. Keep decoder logic combinational.
10. Regression test before every change.

Bank 0 - Address Switches

Switch	Address	Hex
SW1	A15	8000
SW2	A14	4000
SW3	A13	2000
SW4	A12	1000
SW5	A11	0800
SW6	A10	0400
SW7	A9	0200
SW8	A8	0100

Bank 1 - Address Switches

Switch	Address	Hex
SW1	A7	0080
SW2	A6	0040
SW3	A5	0020
SW4	A4	0010
SW5	A3	0008
SW6	A2	0004

SW7	A1	0002
SW8	A0	0001

Bank 2 Controls

SW1 = JP1 VCC path (decode enable)
 SW2 = JP1 CLK path (decode enable)
 SW3 = Auto/Manual
 SW4 = Future auto-run
 SW5-SW8 Reserved

Known Good Tests

1. All switches OFF: Display 0000, decoder disabled, no chip-select LEDs.
2. Bank 2 SW1 ON: Display 0000, decoder enabled, RAM LED on.

Explain Like You're 8

Think of the FPGA as a mail carrier. The address switches make a house number. The 74LS138 decides which neighborhood the house is in. Then the FPGA turns on one LED to show which device (RAM, ROM, UART, or ACIA) is being visited. We always test everything before changing anything.